



PythonX Bootcamp

Python Foundations for Data Science

Final Project

Total Marks: 30

Instructions

- Write clean, well-structured Python code with proper comments.
- Use meaningful variable names and follow good programming practices.
- Perform all required steps using appropriate data science techniques.
- Submit your work as a **Jupyter Notebook (.ipynb)** or **Google Colab notebook**.
- Ensure your analysis is clear, well-organized, and reproducible.

Dataset: Use the following dataset for this project:

<https://www.kaggle.com/datasets/colewelkins/cardiovascular-disease>

Project Tasks & Mark Distribution

1. Data Loading & Understanding (5 Marks)

- Load the dataset into a pandas DataFrame.
- Display basic information (shape, data types, summary statistics).
- Check and report missing values.
- Perform initial Exploratory Data Analysis (EDA) to understand feature distributions.

2. Data Preprocessing (7 Marks)

- Handle missing values appropriately (if any).
- Encode categorical variables (if applicable).
- Perform feature scaling where necessary.

- Select and prepare features for modeling.

3. Model Implementation (7 Marks)

- Build and train the following machine learning models (Just one):
 - Logistic Regression or
 - Decision Tree Classifier
- Split the dataset into training and testing sets.
- Train both models using the training data.

4. Evaluation Metrics (6 Marks)

Evaluate both models using the following metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC Score

Compare the performance of both models and briefly interpret the results.

5. Visualization & Reporting (5 Marks)

Include the following visualizations:

- Confusion Matrix (for both models)
- ROC Curve comparison
- Feature Importance analysis (for Decision Tree model)

Final Note: Your notebook should be well-documented with explanations for each step, making it easy to understand your workflow and results.